Complete Chapter 3 Practice Assessment

1. Suppose $Q = 30.8(0.751)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

2. (a) The annual inflation rate is 3.8% per year. If a movie ticket costs \$9.00 dollars today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

   (b) According to your formula, how much will a movie ticket cost in 30 years?

3. A population has size 8000 at time $t = 0$, with t in years.

   (a) If the population decreases by 125 people per year, find a formula for $P(t)$, the population at time t.

   (b) If the population decreases by 6% per year, find a formula for $P(t)$, the population at time t.

4. If $b > 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to -\infty$? If there is no horizontal asymptote, say so.

5. Find the long term behavior of each of the following functions.

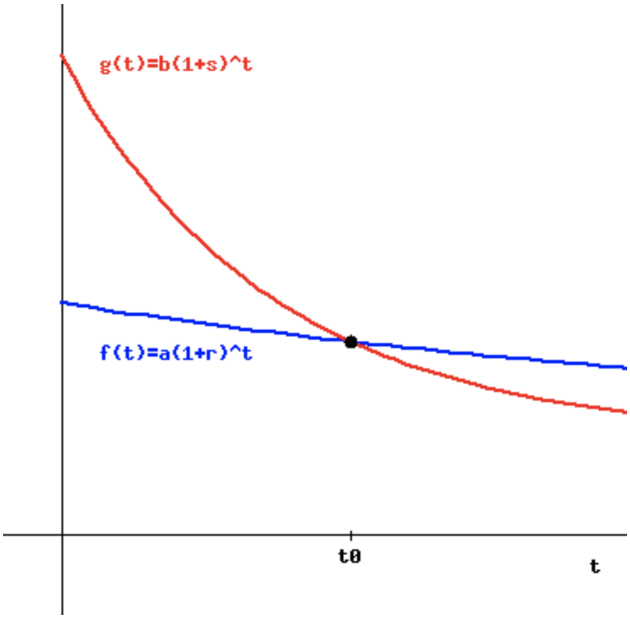
   (a) As $x \to \infty$, $18(0.8)^x \to$

   (b) As $t \to -\infty$, $9(2.2)^t \to$

   (c) As $t \to \infty$, $0.6(2 + (0.1)^t) \to$

   (d) As $t \to -\infty$, $11e^{0.14t} \to$

6. Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1+r)^t$ and $g(t) = b(1+s)^t$. If the value of r is increased and the values of a, b, and s remain the same, does the value of t_0 increase, decrease or stay the same?



7. Certain radioactive material decays in such a way that the mass remaining after t is given by the function

$$m(t) = 175e^{-0.015t}$$

   where $m(t)$ is measured in grams.

(a) Find the mass at time $t = 0$.

(b) How much of the mass remains after 15 years?

8. The graph of the function $f(x) = e^{-x} - 4$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations?

   (a) What are the transformation? Does the order in which you perform the transformations matter? If so, state the correct order.

   (b) State the domain and range of $f(x)$.

9. Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) $\ln 4 = x$

   (b) $\ln x = 3$

10. Evaluate the following expressions. Your answers must be exact and in simplest form.

    (a) $\ln(e^9)$

    (b) $e^{\ln 2}$

    (c) $e^{\ln \sqrt{3}}$

    (d) $\ln\left(\frac{1}{e^2}\right)$

11. Find the exact solution of the exponential equation $e^{2x+1} = 18$.

12. Find the exact solution of the logarithmic equation $6 - \ln(5 - x) = 0$.

13. Find the exact solution of the following equations.

    (a) $3^x = 38$

    (b) $6 \cdot 4^{4x-4} = 65$

    (c) $11 + 5^{5x} = 16$

14. The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 210 degrees Fahrenheit when freshly poured, and 1.5 minutes later has cooled to 195 degrees in a room at 78 degrees.

    (a) Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

    (b) Determine when the coffee reaches a temperature of 155 degrees.